EE5203 L04 Practice Worksheet

Pointers, bits, masks, and CMSIS registers

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Expected time: 30-45 minutes **Policy:** Attempt every question before consulting the worked solutions. All register questions can — and should — be answered on paper first.

Part A - Pointers and structures

1. Trace by hand and give every final value:

```c
/* (a) */                       /* (b) */
int a = 3;                      int m = 10;
int *q = &a;                    int n = 20;
                                int *p = &m;
*q = *q * 4;
a = a + 1;                      *p = *p + n;
                                p = &n;
                                *p = 0;
```

2. `struct Point { int x; int y; };` and `struct Point pt = {4, 7};` `struct Point *pp = &pt;` — for each expression, state its value or say it is invalid:

   a. `pt.x`
   b. `pp->y`
   c. `(*pp).x`
   d. `*pp.y`

3. Explain, in one sentence each, why the device header defines `GPIOA` as a **pointer** to a structure rather than a structure variable, and what `volatile` on each member guarantees.

Part B - Binary and hexadecimal

4. Convert:

   a. `0x0C00` to binary (16 bits) — which bits are set?
   b. `0x00200000` to the single set bit's index.
   c. binary `0000 0100 0001 0001` to hex.
   d. `1U << 13` to hex.

5. GPIOA's `MODER` reads `0xA8000400`. Which pins are **not** in mode `00`, and what is the mode of PA5? (Pin `n` owns bits `2n+1:2n`.)

Part C - Masks

6. Write one statement each (operate on `uint32_t value`):

   a. set bit 12;
   b. clear bit 3;
   c. toggle bit 0;
   d. test whether bit 9 is set (as an `if` condition);
   e. clear the two-bit field at bits 7:6, then write pattern `10` into it.

7. `value` starts as `0x0000 00F0`. Give the value after **each** statement, applied in order:

```
value |=  (1U << 8);
value &= ~(1U << 5);
value ^=  (1U << 4);
value &= ~(3U << 2);
```

8. A student wants PA3 (mode bits 7:6) as output and writes only `GPIOA->MODER |= (1U << 6);`
   a. For which starting field values does this produce the wrong mode?
   b. Write the correct two-statement sequence.
9. Explain the difference between `if (flags & (1U << 4))` and `if (flags && (1U << 4))` — what does each test, and when do they give different answers?

Part D - CMSIS registers and GPIO

10. Decode `RCC->AHB1ENR |= (1U << 0);` word by word: `RCC`, `->`, `AHB1ENR`, `|=`, `1U << 0`. Why is `|=` essential here rather than `=`?
11. The homework wires an LED to PB0 and a button to PC1. Give the exact statements to:
    a. enable the clocks for GPIOB **and** GPIOC in one statement;
    b. make PB0 an output (field bits 1:0);
    c. make PC1 an input with pull-up (`MODER` bits 3:2 = `00`, `PUPDR` bits 3:2 = `01`);
    d. test in an `if` whether PC1 currently reads low (button pressed), using `GPIOC->IDR`.
12. Why is plain assignment (`=`) correct for `GPIOA->BSRR` but destructive for `GPIOA->MODER`? Name the register property that makes the difference.
13. Write the BSRR statements that drive PA5 high and low, and state the general rule for the reset bit of pin `n`.

Part E - Predict the register

14. After generated initialization, `GPIOA->MODER` reads `0xA8000400`. The lab's Checkpoint B/D sequence then executes:

```
GPIOA->MODER &= ~(3U << 10);   /* observed value? */
GPIOA->MODER |=  (1U << 10);   /* observed value? */
```

    Give both intermediate hex values.
15. In the SFR view you observe `RCC->AHB1ENR = 0x00000085`. Which GPIO ports are clocked? (AHB1ENR bit `n` clocks GPIO port `n`: A=0, B=1, C=2, … H=7.)

Part F - Evidence

16. The LED blinks, but you must prove *"PA5 is configured as an output by my two MODER statements, not by leftover generated init."* Describe a breakpoint plan and the two SFR observations that prove it.
17. Build is clean, LED never lights. Give three distinct register-level explanations and the single SFR observation that eliminates each.

Exit self-assessment

Mark one:

☐ Ready for the L04 lab without additional review
☐ Need to review pointers and `->`
☐ Need to review binary/hex and shifts
☐ Need to review the mask recipes
☐ Need to review the RCC/MODER/BSRR roles